

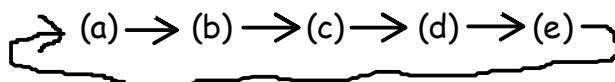
The invertibility theorem

Theorem (invertibility theorem)

Let A be a square matrix. Then the following are equivalent:

- (a) A is invertible
- (b) For every b , the equation
$$Ax = b$$
has a unique solution.
- (c) A is non-singular
- (d) A is row-equivalent to I .
- (e) A is a product of elementary matrices.

Proof We show the following circular chain of implications:



We have already shown most of the implications:

(a) $\rightarrow$ (b) see Thm. 4, last lecture

(b) $\rightarrow$ (c) if A satisfies (b) then A is by definition non-singular

(c) $\rightarrow$ (d) ???

(d) $\rightarrow$ (e) see Thm. 2, last lecture

(e) $\rightarrow$ (a) holds because elementary matrices are invertible and a product of invertible matrices is always invertible.

It remains to show (c) $\rightarrow$ (d).

Let A be any non-singular square matrix. A is row equivalent to a matrix U in row echelon form (any matrix is (Gauss reduction)).

Claim $\forall j, u_{jj} \neq 0$

Proof of the claim

To get a contradiction, suppose that for some $i, u_{ii} = 0$.

If there is a pivot in row i then it must be in a column to the right of column i .
Hence column i has no pivot.

It follows that $U\mathbf{x} = \mathbf{0}$ has non-trivial solutions (because exists a free variable).
Because A and U are row equivalent, it follows that also $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions.

This is a contradiction to the assumption that A is non-singular. □ (Claim)

By Thm. 3 from the last lecture, U is row equivalent to I . Since A and U are also row equivalent, we get (by transitivity of row equivalence) that A is row equivalent to I .

This finishes $(c) \rightarrow (d)$.

This completes the proof of the invertibility theorem.

□ (Invertibility
theorem)

Note The invertibility theorem tells us also for instance that NOT (c) implies NOT (d), i.e., if A is singular then A is not row-equivalent to I .

Determinants

We use a recursive definition here.

We define the determinant of $n \times n$ matrices based on determinants of $(n-1) \times (n-1)$ matrices.

For instance, for 3×3 matrices, we define the determinant as follows:

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

stands for the
determinant of the matrix

$$\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}$$

To define the determinant of $n \times n$ matrices, we first need the following definition:

Definition Let A be an $n \times n$ matrix. Then we define

- (a) $M_{ij} = (i,j)\text{th } \underline{\text{minor}}$ of A
 $=$ determinant of the matrix obtained from A by deleting row i and column j
- (b) $C_{ij} = (i,j)\text{th } \underline{\text{cofactor}}$ of A
 $= (-1)^{i+j} M_{ij}$

Now we can define the determinant of $n \times n$ matrices based on its cofactors (and hence on determinants of $(n-1) \times (n-1)$ matrices):

Definition Let A be any $n \times n$ matrix. Then

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

cofactors of A

This is called "expansion of $\det(A)$ along the first row".

Example

$$\begin{vmatrix} -1 & 2 & 3 \\ 6 & -5 & 4 \\ -2 & 3 & 1 \end{vmatrix} = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

(Note: A blue 'A' is written next to the first row of the determinant.)

$$= (-1) \cdot (-1)^{1+1} \begin{vmatrix} -5 & 4 \\ 3 & 1 \end{vmatrix} + 2 \cdot (-1)^{1+2} \begin{vmatrix} 6 & 4 \\ -2 & 1 \end{vmatrix} + 3 \cdot (-1)^{1+3} \begin{vmatrix} 6 & -5 \\ -2 & 3 \end{vmatrix}$$

(Note: The first minor is labeled -17, the second is labeled 14, and the third is labeled C₁₃ with a value of 8.)

$$= 17 - 2 \cdot 14 + 3 \cdot 8 = \underline{\underline{13}}$$

Theorem 1 We can also expand along any other row or along any column.

Example (expansion along column 2)

$$\begin{vmatrix} -1 & 2 & 3 \\ 6 & -5 & 4 \\ -2 & 3 & 1 \end{vmatrix} = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}$$
$$= 2 \cdot (-1)^{1+2} \begin{vmatrix} 6 & 4 \\ -2 & 1 \end{vmatrix} + (-5) \cdot (-1)^{2+2} \begin{vmatrix} -1 & 3 \\ -2 & 1 \end{vmatrix} + 3 \cdot (-1)^{3+2} \begin{vmatrix} -1 & 3 \\ 6 & 4 \end{vmatrix}$$

(Note: The first minor is labeled 14, the second is labeled 5, and the third is labeled -22.)

$$= -28 - 25 + 66 = \underline{\underline{13}}$$

(Note: A red arrow points from the word 'equal' to the second term in the expansion.)